Grade 8 Name____________________
Unit 14 Date____________________
Lesson 4 Practice Period___________________

As part of a game, a fair six-sided die is rolled and a marble is randomly chosen from a bag containing one red marble, one blue marble, one green marble, and one orange marble. A representation of the marbles is shown.

1. Create a list, table, or tree diagram to determine the sample space.

2. Find the theoretical probability of rolling a 2 and the orange marble is randomly chosen.

P(2, Orange) =

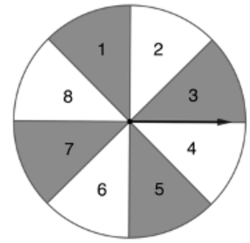
3. Find the theoretical probability that the die *does not* land on a 3 and the marble randomly chosen is red.

P(not 3, red) =

4. What is the theoretical probability that the die lands on an odd number and the marble randomly chosen is a primary color (red or blue)?

P(odd number, primary color) =

In a two-step experiment a fair spinner is spun and a card is randomly picked from a standard deck of cards with 52 cards in four suits - hearts, diamonds, spades, and clubs. The spinner is shown.



5. What is the theoretical probability of landing on 5 on the fair spinner, $\mathrm{P}(5)$?

6. What is the theoretical probability of picking a card with hearts from the deck of cards, $\mathrm{P}(\text{hearts})$?

7. What is the theoretical probability of landing on 5 then picking a card with hearts from the deck, $\mathrm{P}(5, \text{hearts})$?

In a repeated experiment, a card is randomly chosen from a standard deck of cards twice, with replacement.

8. What is the probability of picking a diamond for the first pick, $\mathrm{P}(\text{diamond})$?

9. What is the probability of picking a diamond both times, with replacement, $\mathrm{P}(\text{all diamonds})$?

10. What is the probability of picking a spade both times, with replacement, $\mathrm{P}(\text{all spades})$?